Scientific Reports

https://doi.org/10.1038/s41598-026-48490-7

Article in Press

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Received: 22 December 2025

Accepted: 8 April 2026

Published online: 28 April 2026

Cite this article as: Nisha B.M., Selvakumar J. & Routray S. Machine learning enabled unique chip feature extraction approach for semicustom delay variant PUF. *Sci Rep* (2026). https://doi.org/10.1038/s41598-026-48490-7

B. Muthu Nisha, J. Selvakumar & Soumyaranjan Routray

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Machine Learning Enabled Unique Chip Feature Extraction Approach for Semicustom Delay Variant PUF

Muthu Nisha B[1], Selvakumar J[2]*, Soumyaranjan Routray[3]

[1,2,3] Department of ECE, SRM Institute of Science and Technology, Chennai 603203, India

Corresponding author: Selvakumar J (e-mail: selvakuj@srmist.edu.in).

ABSTRACT The article presents a chip feature extraction approach to create unique and unclonable hardware fingerprints to increase reliability in a hardware security system. A static D flip-flop-based delay counter is used as a delay unit to propose a lightweight PUF in place of the Configurable Ring Oscillators (CROs) in the existing PUF. Initially, the design is simulated and synthesized using three different Complementary Metal-Oxide-Semiconductor (CMOS) technologies to study process variations. Its reliability is ensured through Multi-Mode Multi-Corner (MMMC) analysis. The circuit stands out with its +0.004 ps slack timing and operates stably within a temperature range of 0°C to 125°C. It is quite interesting to observe that it works efficiently within a voltage range of 1.62V to 1.98V. Overall, 155 unique data paths are resolved and carried out by Static Timing Analysis (STA), which ensures the uniqueness of the core design. Furthermore, the results of Machine Learning techniques and the NIST statistical test suite confirm the proposed structure's randomness strength. A comparison of this work with recent findings indicates the specific advantages of the implemented method. The study classifies the proposed Semicustom Delay Variant PUF (SDV PUF) as either strong or weak by adhering to the security requirements.

INDEX TERMS Static D-flip flop, Unique fingerprint, Hardware Security, MMMC, Static Timing Analysis

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I. INTRODUCTION

Integrated circuits (ICs) are becoming more susceptible to copying and unapproved reuse across the supply chain due to the quick globalization of semiconductor manufacturing. The performance, dependability, and security of critical applications such as defence, automotive, and Internet of Things (IoT) devices can be negatively impacted by counterfeit integrated circuits. [1]. PUF, offers a hardware-intrinsic approach in both inexpensive and efficient, generating unique and unpredictable outcomes. Because PUF is influenced by variations in manufacturing processes [2]. PUF technologies facilitate lightweight authentication and safeguard against IC counterfeiting [3]. Although PUFs have many advantages in hardware security, they also have some disadvantages to be considered in terms of resource usage, particularly in systems with limited resources [4],[5]. Addressing these disadvantages is crucial for effective hardware deployment [6]. In PUF categories, one of the silicon delay-based PUFs presently in use is the CRO PUF, which is distinguished from other designs by its simplicity, ease of deployment, and immense CRP space [7]. However, the expanded area utilization due to the doubling of Ring Oscillators (ROs) limits its efficiency for small devices [8].

In this work, the literature review emphasizes how the hardware structure of a configurable system and the inherent process variations play a major role in creating unclonability. In prior literature, numerous configurations and approaches have been proposed to improve the CRO PUF circuits and the approaches for chip features extraction to obtain uniqueness of PUF. Cui et al. presented an hn-CRO PUF that allows for dynamic selection of the number of stages and individual

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inverters in each RO by integrating CMOS inverters as delay elements with RRAM cells acting as programmable switches. By replacing traditional MUX-based configurations, this method drastically lowers hardware overhead [9]. Ram et al. conferred CRO as CMOS inverters with dual voltage control signals to accelerate or retard ageing effects, enhancing both reliability and detection speed. Implemented in 90 nm CMOS technology, the CRO PUF and sensor prove high response stability, area, and power efficiency, and suitability for secure and low-footprint IC applications [10]. Deng et al. deliberated that two hybrid logic gates are used per stage in the adjustable RO PUF for ASICs, which reduces power and space usage. Identical ROs with different frequencies produce distinct and reliable output bits. High uniformity, uniqueness, and dependable performance are confirmed across a range of environmental conditions through experimental validation on a customized PCB [11]. Suresh et al. and Chen et al. considered Tri-gate CMOS PUF cells, which are lightweight delay units that are configured using a 4-bit challenge. However, post-processing is challenging due to their biased and unstable bits [12,13]. Taneja et al. used a Time-to-Digital Converter (TDC) to convert the pulse width into a random digital output and accumulated jitter in ROs as a dynamic source of entropy [14].

Korona et al. sequentially produced response bits by using a Linear Feedback Shift Register (LFSR) as a scrambler and reducing the number of ROs. Nevertheless, the design performed poorly in terms of distinctiveness [15]. Kareem et al. demonstrated a 3-bit challenge-configured XOR-based delay element; yet, the approach relies on shrinking the challenge size to conserve space, which restricts scalability [16]. Wang et al. developed a low-power and incredibly dependable PUF on an ASIC designed for IoT security applications. It is created on a 100 MHz Xilinx ZYNQ 7000 FPGA, and the design is subsequently produced in Cadence Innovus 180 nm CMOS technology. Using reliability, uniqueness, and uniformity measures, the suggested XNOR–XOR RO PUF with a tunable frequency structure achieved 91.92%, 49.62%, and 50.19%, respectively [17]. Pratihar et al. utilized the propagation delay of the buffer and the oscillatory metastability of the Transition Effect Ring Oscillator (TERO) cell to elicit distinct responses. The signal oscillates until it stabilizes in the two identical RO branches that make up the TERO structure. Utilizing the higher inherent delays compared to ASIC designs, the suggested PUF is developed and evaluated on an FPGA platform. Additionally, theoretical IntraHD (intra-response Hamming distance) is used to investigate response stability [18]. Rabiei et al. presented the MCRO-PUF, a modified crossover RO-PUF architecture formed to significantly enlarge the CRP space while improving unpredictability [19]. Wang et al. deliberated the Path-RO approach, which develops on-chip critical path delay measurements to capture process variations. This method influences path delay differences instead of traditional frequency comparison [20]. Hatti et al. offered an enhanced PUF architecture implemented on an FPGA, aiming to improve reliability and resource efficiency. This design validates the feasibility of applying PUFs on programmable hardware [21]. Chen et al., Ni et al., and Yuan et al. deployed their design in Computer-Aided Design (CAD) tools to execute their PUF circuit designs and assess uniqueness based on variances in data path delay [22],[23],[24]. To the best of our knowledge, previous literature lacks in reporting a semicustom design approach with physical layout, geometry, placement, and routing to show chip layout distinctiveness for PUF uniqueness.

These observations inspired the present study of a semicustom design approach. This work contributes a static D-flip-flop-based delay counter as a delay unit instead of a CRO. Therefore, the chip feature investigation has been performed using MMMC analysis and STA to find the uniqueness of PUF. This work uses the Cadence Electronic Design Automation (EDA) tool. The influence of PUF's data path towards hardware unified security is examined and assessed in this study. This article is arranged as follows: Section II discusses the proposed design structure of SDV PUF. Section III explains the process variation using functional simulation and synthesis using different semiconductor technology nodes and delay unit configuration. Section IV deliberates the semicustom design stages. Section V analyses the clock tree synthesis and routing. Section VI

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reports the STA. Section VII compares the performance of the proposed approach with prior literature. Section VIII concludes the paper.

II. PROPOSED SDV PUF: STRUCTURE AND ITS OPERATION

A. DELAY STAGE CONFIGURED BY STATIC D-FLIP FLOP BASED DELAY COUNTERS

This study offers a lightweight PUF design that takes advantage of several built-in variation sources in the existing CRO architecture and substitutes a single configurable delay counter bank for the conventional CRO bank [25]. According to the ISO/IEC 20897-1 (International Organization for Standardization/ International Electrotechnical Commission), a static entropy source is a source of randomness that has been prepared before usage. Motivated by this, the current study adopts static D flip-flop-based delay counters, which have pre-existing, fixed randomization. It can be appropriate for PUF operation. This is possible by manufacturing-induced delay variations [26] [27]. Furthermore, it offers stable data retention without refresh, strong noise immunity, and reliable operation under process and environmental variations [28],[29]. These features help ensure consistent and repeatable PUF responses. Figure 1 shows a hierarchy of modules in top-down approach using verilog for the proposed structure of PUF.

The proposed structure has a 4-bit configurable delay counter. It is initiated when the enable signal is high and the reset signal is low. This delay counter produces a periodic toggling output signal every threshold clock cycles. The output has16 delay control logic (2^4), which is fed into two 3x8 multiplexers. Based on the 6 challenge bits, split into selection bits 1 and selection bits 2. These selection lines make the multiplexer select a specific signal to pass. The counter1 and counter2 gather the number of toggles over time. Finally, the comparator evaluates the counts and produces a response bit.

This proposed work is validated from the semi-custom design aspect. We highlight design validation metrics such as power, area, timing, clock tree synthesis, and static timing analysis to extract the uniqueness of this PUF design.

In the proposed structure, delay counters are used to pose delays. Each counter has an initial delay (Δ_{int}), which adds a slight variation. The threshold for each counter (T_k) is estimated by adding a base value of 5 to the Δ_{int}. This can be formulated as [30] follows in Equation 1.

$$T_k = 5 + \Delta int \quad (1)$$

This proves that each counter executes uniquely, leading to the overall variability needed for the system. Each counter improves its delay value on each clock cycle. The counter value increases by 1 until it attains its threshold, at which the output transition between 0 and 1. The counter updation responds as expressed [31] in Equation 2.

$$dc_k(\Delta+1) = \min (dc_k(\Delta)+1, T_k) \quad (2)$$

Where dc_k is the data point of the delay counter for the k-th counter at time Δ. The counter counts up to the threshold and then switches its output, it is informing that a delay succeeding the threshold. When the delay counter reaches its threshold, the output of the counter changes its state. This means the counter's output varies between 0 and 1 after a certain number of clock cycles, creating a periodic delay signal. This can be stated [32] in Equation 3

$$O_k(\Delta) = \sim O_k(\Delta\text{-}1) \text{ when } dc_k(\Delta) = T_k \quad (3)$$

Where $O_k(\Delta)$ is the output of the k-th counter at time Δ. $O_k(\Delta\text{-}1)$ is the output of the k-th counter at the preceding time step. $dc_k(\Delta)$ is the value of the delay counter for the k-th counter at time Δ.

The design uses multiplexers to select the outputs of the counters. Based on the selection signals (selection bits 1 and selection bits 2), one of the outputs from each set of counters is selected.

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The multiplexers act as selectors, confirming that only one output from each counter set is passed on. The selection is based on [33], in the following Equation 4

$$\{m1out, m2out\} = \{din1[selection\ bits\ 1], din2\ [selection\ bits\ 2]\} \quad (4)$$

Where m1out, m2out is the selected output from the multiplexer. din1,din2 are the inputs to the multiplexer. Selection bits are the selection signal that chooses which input to pass through. The multiplexer picks one of the available outputs from the counters based on the selection signal. The counters as counter1 and counter2, accumulate the selected values from the multiplexers. Whenever the enable signal is active, each counter is updated by adding the respective multiplexer output to its current value. This can be articulated as [34] follows in Equation 5

$$\{C1(\Delta+1), C2(\Delta+1)\} = \{C1(\Delta) + m1out(\Delta), C2(\Delta) + m2out(\Delta)\} \text{ if enable}=1 \quad (5)$$

Where C1 (Δ) and C2 (Δ) are the current values of counter1 and counter2 at time Δ.If the reset signal is active, both counter1 and counter2 are reset to zero. It can be expressed as [35] follows in Equation 6.

$$C1(\Delta), C2(\Delta)\} = \{0\} \quad \text{if reset}=1 \quad (6)$$

The comparator compares counter1 and counter2. If counter1 is greater than counter2, the response bit is set to 1; otherwise, it is set to 0. Response bit ($B_{response}$) can be defined by Equation 7.

$$B_{response} = \{\ 1,\ if\ \ counter1 > counter2\ \ 0,\ if\ \ \ counter1 \leq counter2 \quad (7)$$

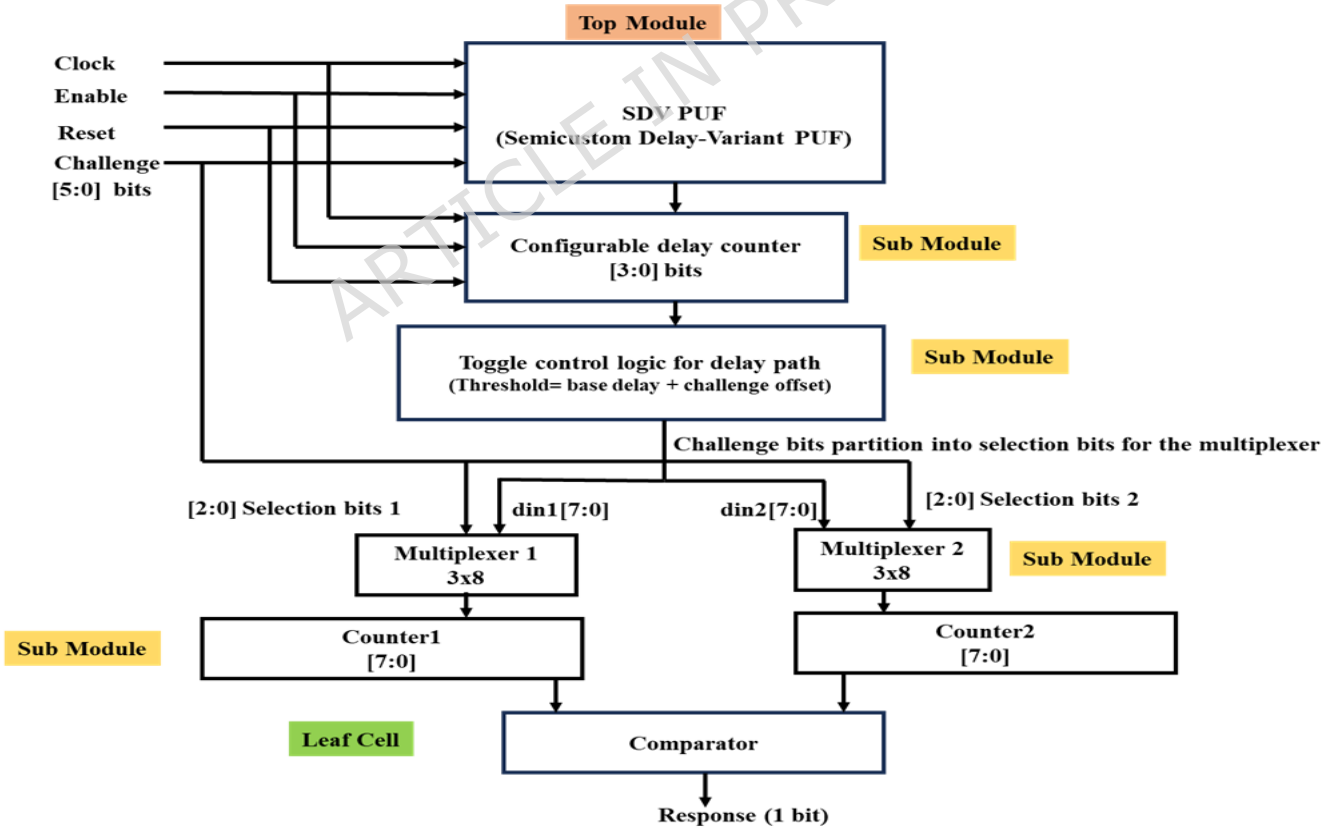


Figure 1. Hierarchy of top-down approach using verilog for proposed PUF.

B. HARDWARE STRUCTURE

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Figure 2 is a basic structure of the CRO PUF with delay units [36]. Those delay units are constructed using static D flip-flop module, positive-edge triggered, and have synchronous active-high enable, as shown in Figure 3. The delay unit is configured by a 6-bit challenge, and the overall structure of the proposed design is illustrated in Figure 4 (a) and (b).

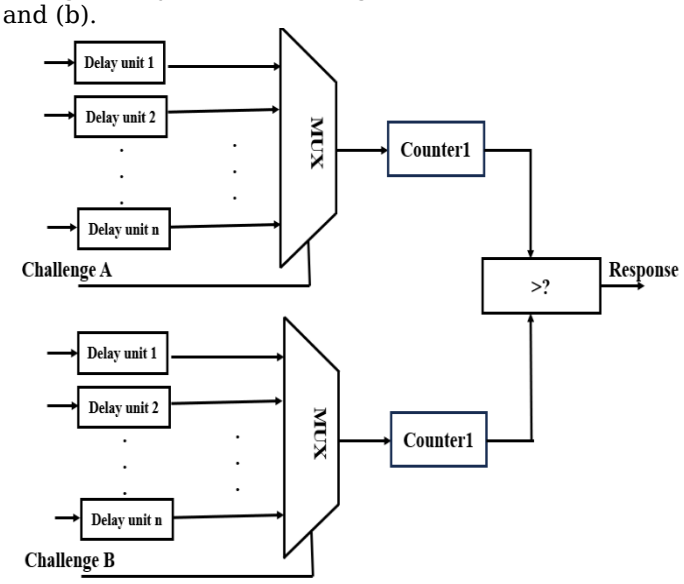


Figure 2. The basic structure of the CRO PUF, where RO is configured through delay units.

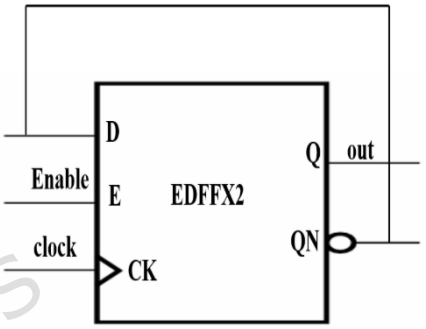


Figure 3. The structure of the Static D Flip-Flop entropy source.

The proposed PUF controls 8-bit multiplexers in two parallel delay routes, each of which controlled by static D flip-flop-based (EDFFX2) delay counters, using a 6-bit challenge input. With its data input (D), clock input (CK), and enable (E), each EDFF generates outputs Q and QN. When several EDFFs cascade together, they create delay counters, whose propagation delays build up over time. A comparator assesses the total delays from both pathways to provide the PUF response, while the 6-bit challenge chooses particular counter outputs to create distinct delay configurations. There are 64 potential input combinations with 6-bit challenges, which creates a large challenge-response space.

III. SIMULATION SETUP AND EXPERIMENTAL RESULTS

The TSMC 180 nm CMOS technology is used to implement chip design. Functional simulation is carried out using Cadence Xcelium, which guarantees the design's logical accuracy before chip layout implementation.

(a)

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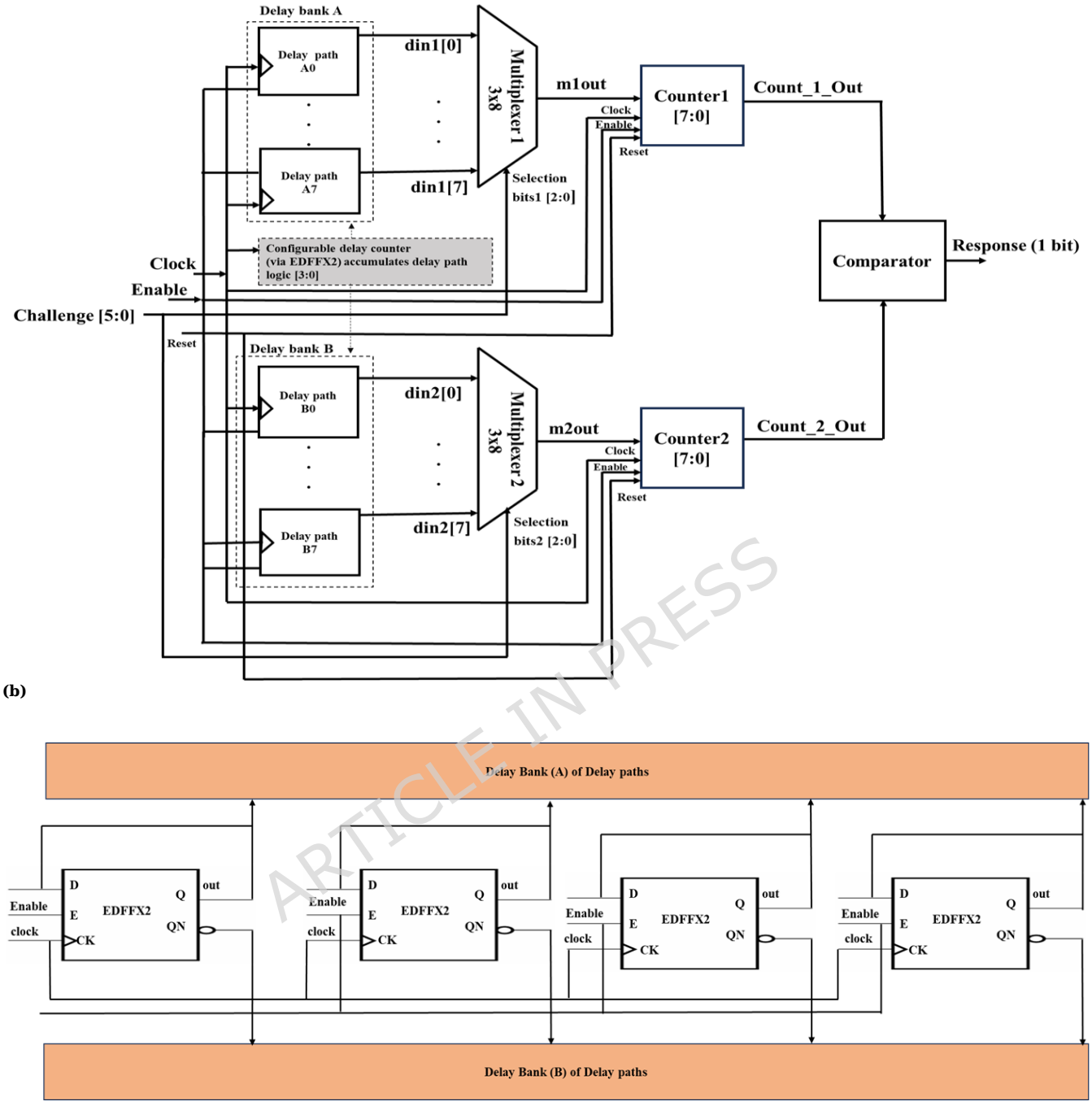


Figure 4. (a) The architecture of the proposed SDV PUF. (b) Configurable delay counter using Static D Flip-Flop

Cadence Genus creates an optimal gate-level netlist through logic synthesis. Floor planning, placement, clock tree synthesis, routing, and layout generation are all done through the Cadence Innovus implementation system. Lastly, STA is performed using Cadence Tempus to assess the delay properties of different instruction activation pathways. For this reason, the proposed PUF is labeled as a semi-custom delay variant PUF (SDV PUF). In this work, the customized circuits are formed and optimized for performance improvement using standard-cell deployment.

A. SYNOPSIS DESIGN CONSTRAINTS

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The synthesis and timing integrity of the proposed PUF are specified through the Synopsis Design Constraints (SDC). These SDCs are outlined in Table I. A clock period of 2 ns is indicated as the primary timing reference for the entire design, imposing the maximum allowable delay per clock cycle. It is about 100 MHz [37]. The rise and fall time transitions, each set to 0.1 ns, signify the clock's transition durations from low to high and high to low states, respectively, confirming accurate edge timing for synchronous elements. An uncertainty margin of 0.01 ns is combined to account for variations in clock arrival times and other environmental factors, acting as a guard band to develop robustness during STA. Additionally, a maximum input delay of 1.0 ns is stated, indicating that the expected maximum time for external input signals to realize the internal logic. Similarly, a maximum output delay of 1.0 ns is set to assure that output signals from the design are generated within the allowable time frame. These constraints together lead to the synthesis and optimization to meet the main performance, timing closure, and functional precision of the proposed PUF.

TABLE I
SYNOPSYS DESIGN CONSTRAINTS

Constraints	**Specification**	**Meaning**
Clock Period	2 ns	Signifies a timing reference to the design
Rise time transition	0.1 ns	Signifies the clock from low to high
Fall time transition	0.1 ns	Signifies the clock from high to low
Uncertainty	0.01 ns	Signifies guard band for timing analysis
Maximum input delay	1.0 ns	Signifies the input signal to propagate
Maximum output delay	1.0 ns	Signifies the output signal to propagate

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B. FUNCTIONAL SIMULATION OF PROPOSED PUF

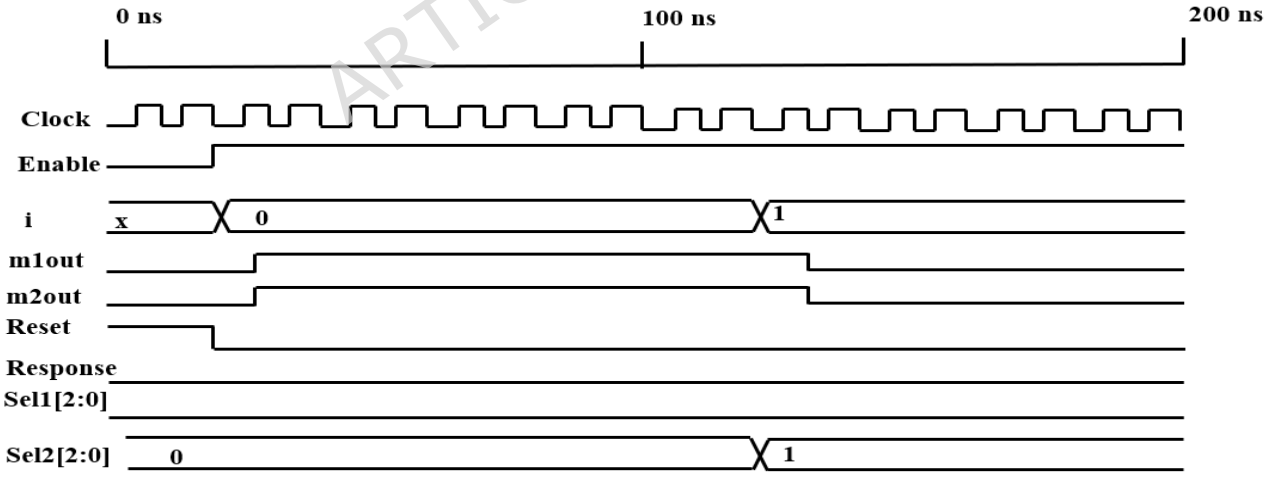


Figure 5. Functional simulation of SDV PUF.

A segment of the simulation timeframe for the proposed PUF can be seen in Figure 5, specifically highlighting the 0 to 200 ns timeframe in which two challenge inputs are employed. While the total simulation duration goes beyond 200 ns to confirm full propagation and stable response capture, based on the time frame reveals the system's functional behavior under two distinct challenges. The PUF is activated via a constant enable signal, and operations are coordinated using a

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continuous clock. The challenge input ‘i transitions from an undefined state to '0' at approximately 10 ns, then to '1' at 100 ns, enabling two validation cycles. Selection signals Sel1[2:0] and Sel2[2:0] target the internal signal paths accordingly. Intermediate outputs (m1out and m2out) expose the differential delay paths characteristic of the PUF's core design. A brief reset pulse at around 40 ns assures proper resetting between valuations. The response stays stable during this interval, verifying the system’s ability to reliably change between consecutive challenge inputs. Through this process, Challenge Response Pairs (CRPs) can be collected for different PUF sizes.

C. PROCESS VOLTAGE TEMPERATURE CONDITIONS

The reliability can be validated for the PUF circuits through PVT conditions. Here, the PVT conditions for three distinct semiconductor technology nodes (180 nm, 90 nm, and 45 nm) beneath slow and fast process corners are shown in Table II. Each technology node indicates the voltage, temperature, and power rails (Rail_VDD and Rail_VSS). Across these conditions, the "Slow" corner represents lower performance with higher power consumption, while the "Fast" corner denotes higher performance with lower power consumption. The operating voltage varies at different semiconductor nodes, with higher nodes typically demanding higher voltages. Also, Temperature conditions change between corners, with the slow corners confirmed at high temperatures (125°C) and the fast corners confirmed at lower temperatures (0°C). The Rail_VDD and Rail_VSS values indicate the supply voltage levels used during operation, which are vital for observing the overall power and performance attributes of the semiconductor nodes.

TABLE II
PROCESS VOLTAGE TEMPERATURE CONDITIONS

Description of semiconductor nodes	**PVT Conditions**				
	Process- P	**Voltage-V (Voltage)**	**Temperature-T (^{0}C)**	**Rail_VDD (Voltage)**	**Rail_VSS (Voltage)**
180 nm (fast)	1	1.98	0	1.98	0
180 nm (slow)	1	1.68	125	1.68	0
90 nm (fast)	1	1.1	0	1.1	0
90 nm (slow)	1	0.9	125	0.9	0
45 nm (fast)	1	1.32	0	1.32	0
45 nm (slow)	1	1.08	125	1.08	0

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D. IMPACT OF PROCESS VARIATION

The combined findings of different PUF sizes (8, 16, 32, and 64 configurations) and semiconductor technology nodes (180 nm, 90 nm, and 45 nm, across fast and slow process corners) for the proposed design in terms of area, power, and timing are offered in Table III. As the configuration increase, a consistent rise in both cell count and total area is noted, indicating a relative progress in logic complexity.

The total area increases from 11,878.57 µm² at 8 CROs (180 nm fast) to 96,645.23 µm² at 64 CROs. Power consumption also scales accordingly, increasing from 6.21 nW to 49.27 nW over the configuration. Additionally, designs synthesized on smaller nodes, such as 45 nm, confirm significantly reduced power and area compared to 180 nm and 90 nm technologies. It emphasizes the advantages of advanced technology scaling. In terms of timing, fast process corners consistently provide greater slack, while timing performance lowers with higher CROs and slower corners. Especially, negative slack is encountered at 64-bit under the 180 nm slow corner (-56 ps), showing a potential timing violation. It means the signal reaches after the required time. Negative slack can be the source of functional failure, as flip-flops may hold incorrect data, leading to glitches or wrong outputs. Multiple negative slack paths can make timing closure issues, risking chip verification failure. This can be mitigated through clock tree synthesis in semicustom design flow. These results collectively explain the critical influence of CROs and semiconductor technology node on design scalability, performance, and feasibility.

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Figure 6 demonstrates the variation in design metrics, area, power, and slack, across different PUF size and semiconductor technology nodes under both fast and slow process corners. Figure 6 (a) displays a clear increase in area with rising CROs, as expected due to the increasing complexity of the logic. Among the nodes, 45 nm presents the lowest area footprint across all configurations, while 180 nm always consumes the most cell count, mainly under slow corners. Figure 6(b) represents power consumption trends, indicating a significant rise with increasing PUF size.

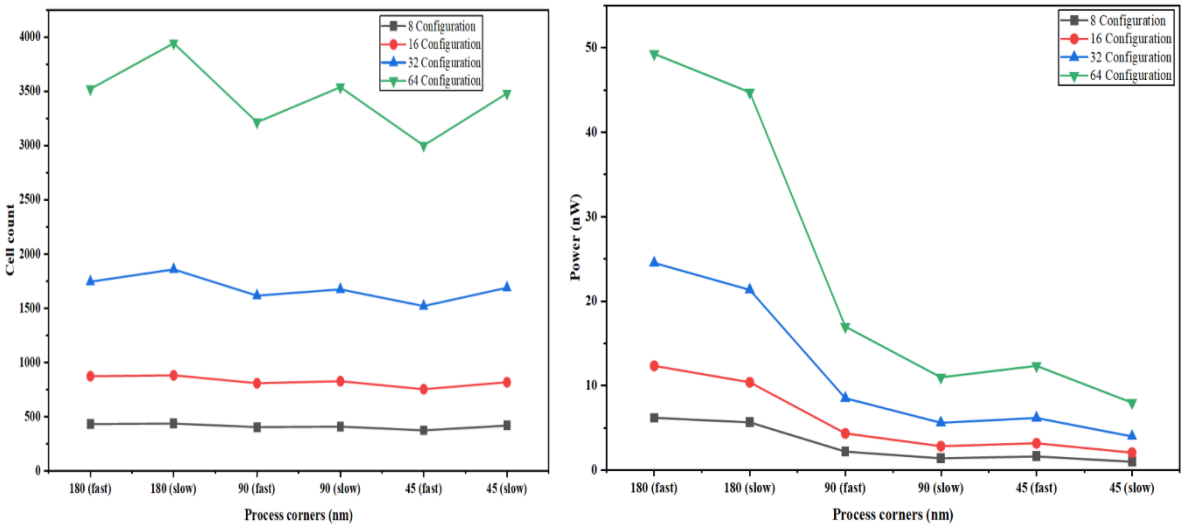
The 180 nm node utilizes the highest power, especially under fast corner conditions, while 45 nm provides the most energy-efficient profile. Figure 6(c) focuses on timing slack behavior, exposing that slack generally decreases as the configuration increases. Particularly, 45 nm (slow) and 90 nm (fast) corners retain positive and higher slack values, resulting in better timing margin. However, the 180 nm (slow) configuration reaches negative slack at 64 configuration, implying a potential timing violation. Overall, the analysis highlights the scalability advantages of different technology nodes.

TABLE III
PERFORMANCE OF THE PROPOSED PUF AT DIVERSE BIT WIDTHS AND TECHNOLOGY SCALES

PUF configuration with delay units	**Semiconductor technology node**	**Area (Cell/Total in µm²)**	**Power (nW)**	**Timing/Slack (ps)**
8 Configuration	180 nm (fast)	434/11878.575	6.210821	707
	180 nm (slow)	438/12457.368	5.67993	2
	90 nm (fast)	406/3582.408	2.21027	1429
	90 nm (slow)	411/3642.960	1.41621	15
	45 nm (fast)	376/0	1.64236	1576
	45 nm (slow)	420/0	1.02374	653
16 Configuration	180 nm (fast)	875/23886.879	12.3704	530
	180 nm (slow)	883/25337.189	10.4217	8
	90 nm (fast)	810/7217.041	4.36637	1264
	90 nm (slow)	828/7404.753	2.85597	8
	45 nm (fast)	754/0	3.20524	1433
	45 nm (slow)	819/0	2.09578	138
32 Configuration	180 nm (fast)	1746/47634.049	24.5377	244
	180 nm (slow)	1859/52720.114	21.3481	0
	90 nm (fast)	1618/14417.431	8.53098	1099
	90 nm (slow)	1676/14976.023	5.63502	0
	45 nm (fast)	1521/0	6.20125	1370
	45 nm (slow)	1690/0	4.00913	144
64 Configuration	180 nm (fast)	3522/96645.227	49.2734	102
	180 nm (slow)	3943/109997.395	44.7331	-56
	90 nm (fast)	3216/28792.476	17.0081	725
	90 nm (slow)	3538/29980.052	11.0096	1
	45 nm (fast)	3002/0	12.3648	1192
	45 nm (slow)	3482/0	8.01220	10

(a) (b)

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(c)

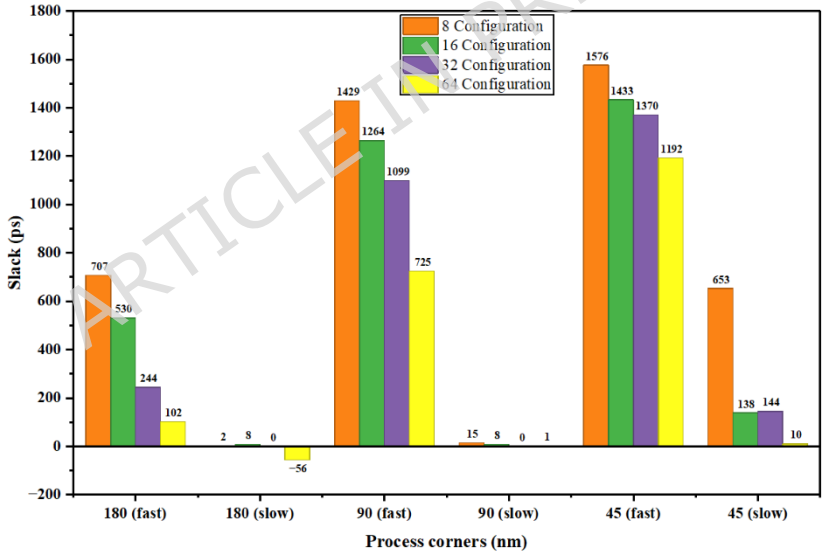


Figure 6. Performance of counter-regulated PUF with process variation (a) Measured Cell count(b) Measured Power (c) Measured slack.

IV. INITIATION OF SEMICUSTOM DESIGN STAGES

The semicustom design of a proposed PUF begins with MMMC analysis. Obviously, it is necessary to integrate the designs of floor planning, power planning, and the placement of standard cells to obtain delay variant PUF [38],[39],[40].

A. DEPLOYMENT OF MMMC ANALYSIS

The MMMC analysis is based on the setup and hold analysis. This is configured for a 180 nm technology logic design for better timing margin. It begins by creating Slow-Slow (SS) and Fast-

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Fast (FF) corners. The maximum timing is using a SS corner for worst-case (setup) analysis, and another for minimum timing using the FF corner for best-case (hold) analysis. The analysis is driven by the SDC, which contains the design's timing and cell instances. Next, two delay corners are defined by maximum delay using the SS corner for setup timing, and minimum delay using the FF corner for hold timing. Analysis perspectives are named as worst and best cases. Each case is associated with the corresponding delay corner. This makes the PUF design perfectly analyzed under appropriate timing conditions for both setup and hold standards [41].

B. FLOOR PLANNING

The core size of the proposed PUF is denoted in Table IV according to the aspect ratio, which is approximately 0.8574, representing a rectangular core shape where the height is less than the width. Both the core and cell utilization factors are set to 0.6999, which means that around 70% of the core area will be loaded by standard cells [42], leaving space for routing and other overhead. The core margins result from the core to the IO boundary, with equal spacing of 8 microns on all sides (left, right, top, and bottom). The die size computation uses maximum IO height, and the floorplan origin is set to the lower left corner. These parameters ensure a well-balanced and routable PUF layout using its design constraints, which are exposed in Figure 7.

TABLE IV
FLOOR PLANNING SIMULATION SETUP

Floor Plan Design Metrics	**Specification Size (microns)**
Aspect ratio (Height/Width)	0.85739
Core utilization	0.6933
Core margin to the right	8
Core margin to the left	8
Core margin to the top	8
Core margin to the bottom	8

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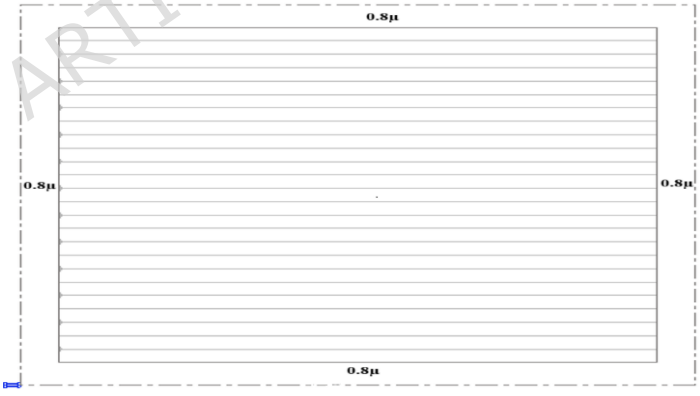


Figure 7. Floor plan segment for the core PUF Visualization

C. POWER PLANNING

Power rings are shaped around the core boundary to deliver power evenly around the core PUF. The formation of power rings is demonstrated in Table V. The power stripes are added across the core to bring power into the center. These structures ensure that power is carried efficiently and uniformly to all parts of the chip. The VDD (power) and VSS (ground) rails are formed around the core boundary of an integrated circuit. The power rings are formed using Metal6 for the top and

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bottom sides (horizontal direction) and Metal5 for the left and right sides (vertical direction), with each metal line having a width of 1.8 microns and a spacing of 0.25 microns between VDD and VSS, as illustrated in Figure 8. The offset is about to center the rings within the routing channel. This shape proves low resistance and efficient power delivery across the chip [43], which is essential for minimizing IR drop (Current-I, Resistance-R drop) and supporting the steady operation of the internal cells.

TABLE V
POWER RING CONFIGURATION SETUP

Power rail setup	Layer	Width (in micron)	Spacing (in micron)
Top	Metal 6	1.8	0.25
Bottom	Metal 6	1.8	0.25
Right	Metal 5	1.8	0.25
Left	Metal 5	1.8	0.25

D. PLACEMENT OF STANDARD CELLS

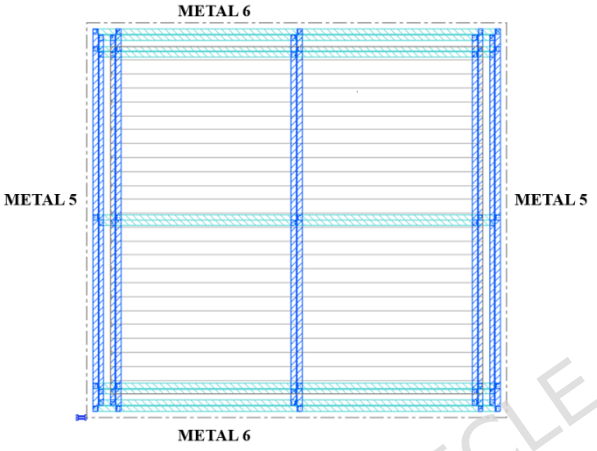


Figure 8. Visual representation of power planning for the core PUF of the placement of standard cells

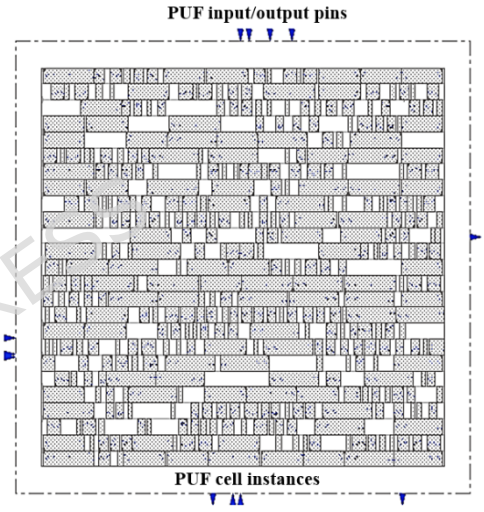


Figure 9. Visual representation

In this layout of Figure 9, the standard cells are organized in a row-based structure where each row has a uniform height similar to that of the cells, permitting efficient routing and power distribution. The cells are placed side by side to achieve area utilization.

V. TIMING RELIABILITY ANALYSIS

A. IMPORTANCE OF CLOCK TREE SYNTHESIS (CTS) AND ROUTING

CTS is a significant process in semicustom design of delay variant PUF that confirms the accurate and effective delivery of clock signals to all sequential elements, with optimized signal routing for maintaining timing integrity [44]. The worst-case and best-case analyses are involved in CTS. The routing is to make sure timing reliability. The best-case analysis confirms hold time under fast process, high voltage, and low temperature settings, whereas the worst-case analysis looks for setup time violations under slow process, low voltage, and high temperature conditions. These estimations under all operating conditions impact clock balancing, skew reduction [45]. It improves the design stability with positive slack and reliability of the proposed PUF.

B. WORST CASE ANALYSIS

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The setup analysis for the various design stages appears in Table VI. In the pre-CTS phase, the PUF design had an area of 416 units, a power of 6.00195 nW, and a Worst Negative Slack (WNS) of -2.976 ps with 112 timing violations. After pre-CTS optimization, the area increased to 465 units, power decreased to 5.37209 nW, and all timing violations cleared with a WNS of +0.002 ps. In the post-CTS phase, area and timing endured the same, while power slightly increased to 5.73208 nW. After routing, the area sustained at 465 units, power increased to 5.82783 nW, and WNS became slightly negative about -0.020 ps with 2 violations. Finally, in post-route optimization, the area increased to 467 units, power to 5.83264 nW, and timing violations are removed with a WNS of +0.004 ps. This non-zero slack is a gain of the proposed architecture. The design reduces systematic bias and confirms that the generated response is dominated by intrinsic process variations, not by deterministic architectural imbalance.

C. CTS

A CTS is deployed to develop the PUF clock network performance. Furthermore, to mitigate the detected negative slack under worst-case slow corner conditions, clock tree synthesis (CTS) is optimized by non-default routing rules, upper metal layers, and high-drive clock buffers. Consequently, wider clock routing and compact resistance lowered clock latency. These measures efficiently improved clock arrival times and restored positive setup slack without changing the functional data paths, confirming robust operation across all PVT corners. The routing width and spacing are expanded for the major metal layers by a non-default rule (2w2s) in the CTS configuration. The clock route is limited to Metal5 and Metal6 to lower delay and congestion. Specific buffer, inverter, and clock-gating cells are assigned to optimize drive strength and power efficiency. The whole place-and-route proposed PUF integrated circuit outline is shown in Figure 10. The colored lines denote various metal layers for routing signals, power, and clocks across the chip. Vertical and horizontal yellow, red tracks indicate power/ground rails and clock distribution networks passed through the core. The dense sections of colored polygons are linked with standard cell placements within defined rows, showing functional blocks like logic gates and flip-flops. The layout also incorporates input/output pins at the border and vias connecting different metal layers. This layout validates the semicustom design finishing point, enhancing design rule compliance and signal routing for fabrication readiness of PUF.

D. BEST CASE ANALYSIS

In the post-CTS phase, the hold analysis (best case) signifies the number of analyzed paths and the minimum delay. Table VII describes the hold analysis for various design stages. Across all stages, the number of paths continued to be stable at 465 cells, enhancing slightly to 467 cells after post-route optimization, implying minimal structural alterations during the final tuning. The cell area confirms a modest growth from 13,185.850 μm^2 to 13,308.926 μm^2. This rise is acknowledged to include additional buffer insertion and gate resizing operations to develop timing and signal integrity during optimization phases. Despite these alterations, the design continued to be highly area-efficient.

TABLE VI
SET UP ANALYSIS

Design Stages	**Set up Analysis (Worst Case)**			
	Area	**Power (nW)**	**Timing**	
			Worst Negative Slack (ps)	**Violating Paths**
Pre CTS	416/11758.824	6.00195	-2.976	112
Pre-CTS Optimization	465/13185.850	5.37209	+0.002	0
Post CTS	465/13185.850	5.73208	+0.002	0
Post Route	465/13275.662	5.82783	-0.020	2

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Post-Route Optimization	467/13308.926	5.83264	+0.004	0

Table VII
HOLD ANALYSIS

Design Stages	**Hold Analysis (Best Case)**			
	Area (cells/μm²)	**Power (nW)**	**Timing**	
			Worst Negative Slack (ps)	**Violating Paths**
Post CTS	465/13185.850	5.73208	+0.054	0
Post-CTS Optimization	465/13185.850	5.82617	+0.054	0
Post Route	465/13185.850	5.82783	+0.054	0
Post-Route Optimization	467/13308.926	5.83264	+0.004	0

Regarding power consumption, a small increase is noted from 5.73208 nW at post-CTS to 5.83264 nW after post-route optimization. This steady increase exhibits the overhead presented by additional logic elements, yet stays within acceptable bounds, demonstrating that power integrity is properly sustained. Significantly, timing results show a regularly positive WNS, with values of +0.054 ps across the initial stages and a final WNS of +0.004 ps at post-optimization. This confirms that the design provides all hold constraints with zero violating paths examined at any stage. The regular absence of violations confirms the robustness of the CTS and routing approaches applied.

VI. UNIQUE CHIP FEATURE EXTRACTION APPROACH

A. ANALYSIS FOR DATA PATH DELAY

The timing summary and path delay distribution for each timing path in the design, as computed by a STA, are displayed in Figure 11. The path histogram reveals the number of timing paths falling into different delay ranges, with the x-axis detailing the safe margin of delay path, and the y-axis detailing the count of paths.

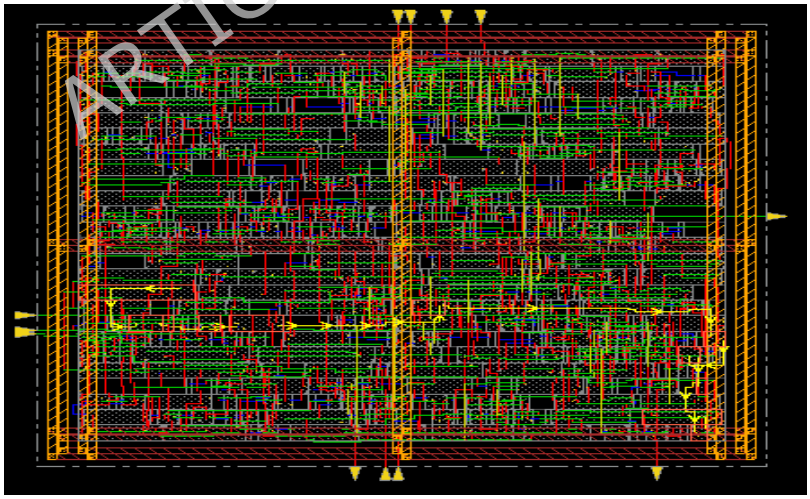


Figure 10. Final layout of the place and routing of the proposed PUF

The distribution details that most paths fall within precise clusters of delay, revealing the variability due to logic depth and routing. this reports that all 155 paths in the design have passed the timing checks, with zero failing paths for complete timing closure. Both WNS and Total Negative Slack

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(TNS) are reported as 0.0000, proving that no path violates timing constraints under worst-case conditions.

This result confirms that the design meets its target clock frequency without any hold or setup violations across 155 timing paths. To validate the timing performance of the PUF circuit, STA is executed across 155 critical paths. The results are measured in terms of slack and required time for each path. As shown in Figure 12, the graph displays the variance in slack and required time for various timing paths in the proposed PUF. Every path has an individual blend of slack and delay values, thus, no two paths act in the same way. This variance confirms that every path has a distinct, unique temporal signature and is mostly the result of process and routing variations. For hardware security applications of this proposed PUF, these unique timing path features show good uniqueness, which guarantees that even under the same conditions, each chip or path generates a unique response.

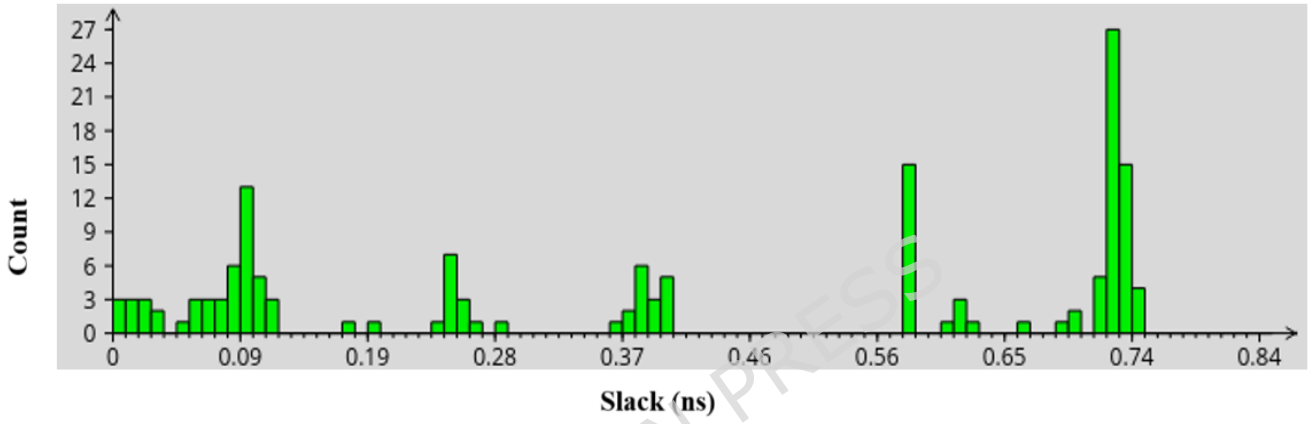
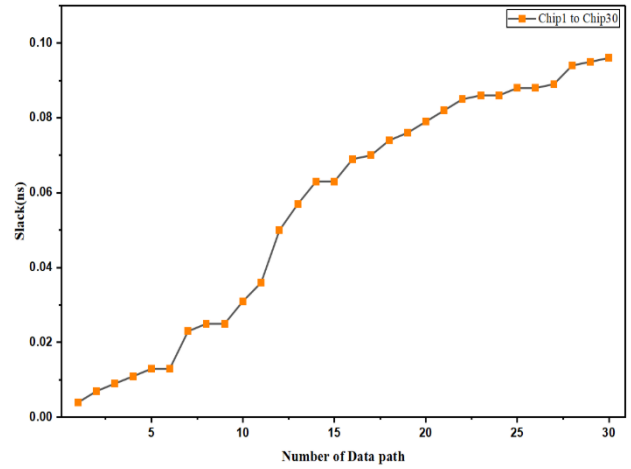


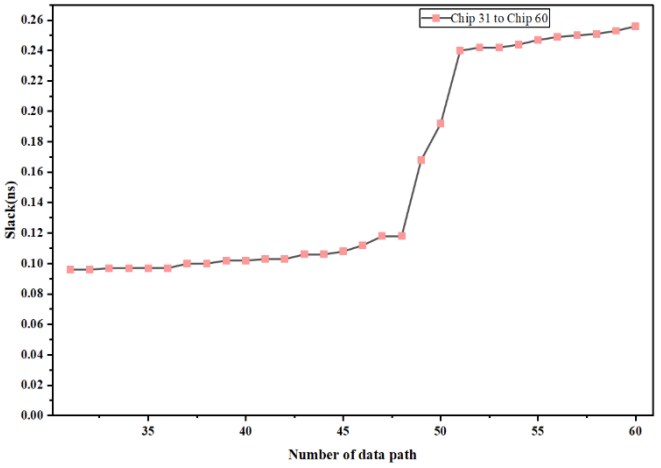
Figure 11. Visual representation of histogram data path distribution in the STA analyzer

Table VIII records the distribution of 155 timing propagation paths grouped by their slack values. The majority of paths (62 paths) fall within the slack range of 0.601–0.750 ps, revealing a healthy timing margin. A smaller number of paths (3 paths) are within the 0.000–0.010 ps range, suggesting they are close to violating timing constraints and may require further optimization. This distribution assists in identifying critical and near-critical paths in the design, supporting targeted timing improvements. The two delay variant paths are captured out of 155 data paths, which are demonstrated using chip layout in Figure 13 and Figure 14. Therefore, this denotes the unique hardware intrinsic signatures through a semicustom approach. Hence, this work introduces Semicustom Delay Variant PUF (SDV PUF).

(a) (b)

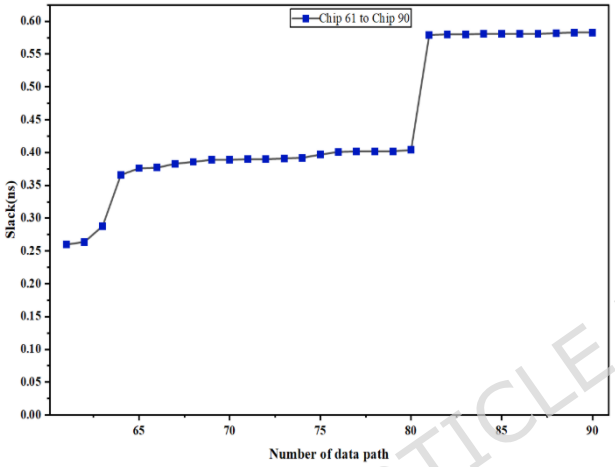
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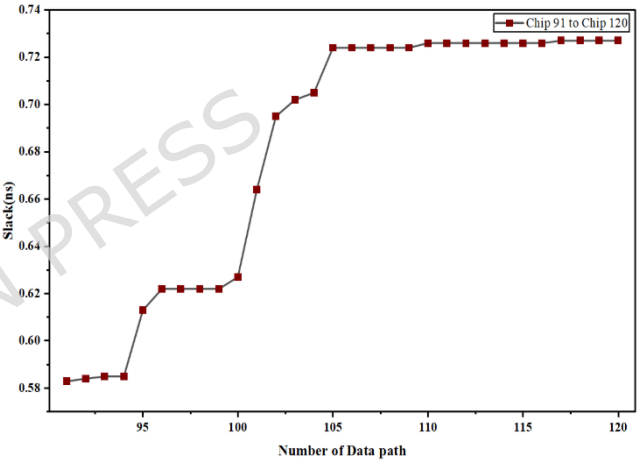




(c)

(d)





(e)

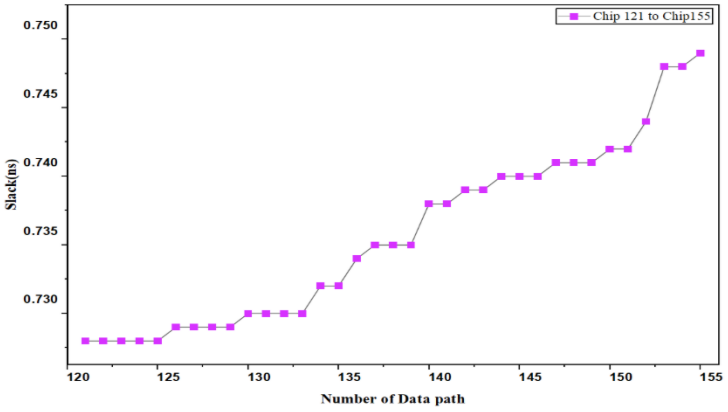


Figure 12. Measurement of slack across data paths (a) Chip 1 to 30 (b) Chip 31 to 60 (c) Chip 61 to 90 (d) Chip 91 to 120 (e) Chip 121 to 155

In the last phase of semi-custom design, after routing, GDS II (Graphic Data System II) is protected for the fabrication phase. This information includes a mixture of encoded data. It signifies

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special binary values, signal names, module references, and other metadata for the fabrication of the proposed PUF.

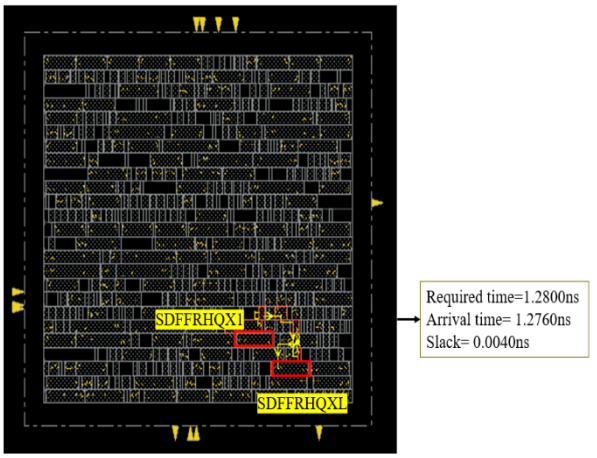


Figure 13. Layout of a unique signature on chip-I out of 155 data paths

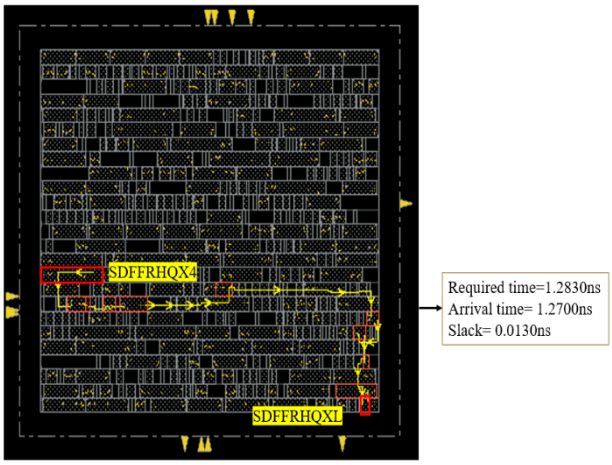


Figure 14. Layout of a unique signature on chip-II out of 155 data paths

TABLE VIII
UNIQUE CHIP VARIATIONS ACROSS SLACK RANGES

Slack Range	Number of unique chip differentiations
0.000-0.010	3
0.011-0.050	9
0.051-0.100	26
0.101-0.300	25
0.301-0.600	31
0.601-0.750	61

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B. MACHINE LEARNING ATTACK

The machine learning (ML) modelling results of the proposed SDV PUF are shown in Table IX for different PUF sizes using a challenge–response pair (CRP) set size of 1000 initially. Numerous well-known machine learning methods were used, such as Long Short-Term Memory (LSTM) networks, Random Forest (RF), XGBoost, Gaussian Mixture Model (GMM), Logistic Regression (LR), Support Vector Machine (SVM) with a sigmoid kernel, K-means, K-means++, and Multilayer Perceptron (MLP).

TABLE IX
RESULTS OF MACHINE LEARNING MODELLING ACCURACY

PUF	ML analysis	CRP Set Size (1K-10K)	Accuracy (%)			
			PUF Size depends on the delay units			
			8 configurations	16 configurations	32 configurations	64 configurations
SDV PUF	LR	1000	71.5	70	64.5	0.645
	SVM sigmoid	1000	38	44	51	0.37
	GMM	1000	59	56	55.5	0.645
	K-means	1000	56	56	51	0.54
	K-means++	1000	0.56	56	51	0.54
	RF	1000	100	100	85	0.885
	XG BOOST	1000	100	100	92	0.905
	MLP	1000	100	100	92.5	0.915

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	LSTM	1000	75.9	66	61	0.824
	LR	10000	68.93	72.46	62.6	64.6
	SVM sigmoid	10000	46.2	57.33	36.53	56.26
	GMM	10000	52	50.8	52.53	50.78
	K-means	10000	55.57	50.8	50.92	50.05
	K-means++	10000	62.03	50.8	50.46	50.43
	RF	10000	68.97	73.8	64.13	64.67
	XG BOOST	10000	65.87	73.8	56.27	64.67
	MLP	10000	68.93	78.73	64.2	64.66
	LSTM	10000	68.5	73.8	61.2	64.67

The findings show that the proposed PUF shows high resistance to linear and probabilistic modeling techniques, with modeling accuracies falling below 70% for traditional learning algorithms, including LR, SVM, GMM, and K-means variations. Conversely, more nonlinear models like RF, XGBOOST, and MLP achieved more accuracy, often reaching 100%, demonstrating their superior ability to fit intricate delay-based relationships. In spite of this, the temporal dependency-learning LSTM model obtained relatively reasonable accuracy of 0.759, 0.66, 0.61, and 0.824 for PUF configurations, respectively. While improving CRP set size from 1K to 10K, RF, XGBOOST, and MLP produce lower modeling accuracy. It improves randomness strength with vast CRP size. It is observed that, ML accuracy varies by different PUF configurations. this behavior can be attributed due to the relationship between feature dimensionality, challenge-response density, and noise. This makes it clear that even when assessed using sophisticated neural network-based techniques, the SDV PUF retains an acceptable degree of modeling resistance and unpredictability.

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C. STATISTICS PERFORMANCE

Two popular statistical performance metrics are considered to evaluate the proposed PUF, as described in [46] and [47]: uniformity (ideal value: 50%) and uniqueness (ideal value: 50%). When describing a particular PUF type, we will need the following parameters in order to get quantitative estimates:

- Uniqueness: This statistic calculates how well a PUF can recognize a particular chip from a collection of chips of the same type. If two distinct PUF instances, Pi and Pj, respond to the identical challenge with Ri and Rj, respectively, then

$$\text{Uniqueness}=\frac{2}{m(m-1)}\sum_{i=1}^{m-1}\sum_{j=i+1}^{m}\frac{HD(R_i,R_j)}{m} * 100\% \quad (8)$$

Equation 8 calculates the average Hamming Distance (HD), scaled by the factor $\frac{2}{m(m-1)}$, and divided by the normalization factor m, across all pairs of responses (Ri, Rj) for every combination of i and j.

- Uniformity: According to the definition of uniformity, the response's distribution of l’s and 0’s should be equal. Randomness is another term for it. The response received for the l's distribution.

$$\text{Uniformity} = \frac{1}{m}\sum_{t=1}^{m} R_{i,t} * 100\ \% \quad (9)$$

Uniformity suggests that the response's ls and 0s should be distributed equally. As per Equation 9, a perfect PUF exhibits 50% uniformity. The calculated PUF metrics, uniqueness, and uniformity are derived from simulation CRP data using the above formulas, which are shown in Table X.

TABLE X

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RESULTS OF STATISTICAL PERFORMANCE METRICS

PUF	PUF Size	Uniqueness (PUF1, PUF2)	Uniformity
SDV PUF	8 Configurations	40.10%	34.90%
	16 Configurations	55.92%	37.20%
	32 Configurations	27.98%	42.20%
	64 Configurations	45.30%	55.20%

From Table X, the proposed PUF is mostly based on intentional path asymmetry in place of nonlinear post-processing functions. As a result, a slight bias is formed during the initial configuration of the design. This primary bias propagates through the delay comparison mechanism, causing observable fluctuations in uniqueness and uniformity metrics, mainly for smaller configuration sizes like 8 to 32 delay units.

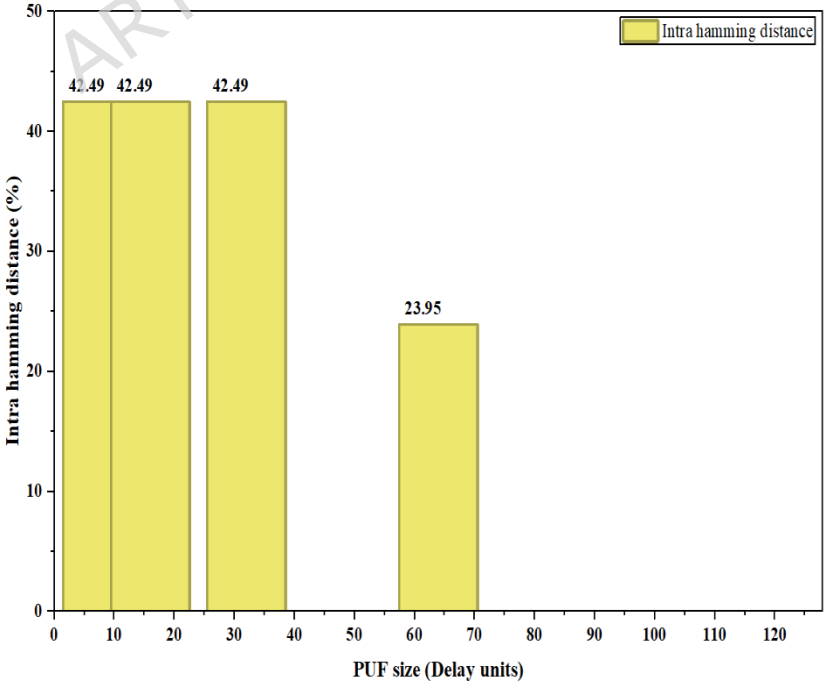
D. INTRA-DIE HAMMING DISTANCE

The stability of PUF responses for the same challenges under identical conditions is evaluated by the intra-die Hamming Distance (HD) using Equation 10 in [48]. Perfect repeatability is indicated when the HD is around 0%. Nevertheless, the HD values in our investigation are 23.95% for the 64 delay unit PUF and 42.49% for the 8, 16, and 32configuration PUFs. The 64 configuration design exhibits better stability because of the average path delays, compared to the smaller PUFs display more fluctuation. There are modifications needed to reach optimal stability levels. The results are shown in Figure 15.

$$\text{IntraHD(R, } R_m) = \frac{1}{m}\sum_{t=1}^{m} r_m - r_{tm} \quad (10)$$

Where R is the response of the proposed PUF, R_m response with other PUF condition, m is the number of bits in the response, r_m is mth bit of the proposed response, r_{tm} is the mth bit of the tested response.

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Figure. 15 Intra-chip stability based on hamming distance

E. NIST SP 800-22 STATISTICAL TEST SUITE

The randomness of the PUF responses is estimated with the NIST SP 800-22 statistical test suite. The bitstream is separated into 50 sequences, each with a length of 993 bits. In this work, 49,650-bit response sequences are obtained from Xcelium console simulations. Before testing, an XOR-based whitening of preprocess is implemented to reduce bias in the raw responses. The experimental outcomes prove that all tests achieved pass rates above the suggested threshold of 0.8, signifying strong statistical randomness. The test results are noted in Table XI.

TABLE XI

NIST RESULTS

Test	**Number of Runs**	**Length**	**Average P-value**	**Pass rate**	**Pass?**
Frequency	50	993	0.5310	1.00	Yes
Runs	50	993	0.4459	1,00	Yes
Longest Run	50	993	0.3961	1.00	Yes
FFT	50	993	0.9486	1.00	Yes
Cumulative Sums	50	993	0.1706	0.86	Yes
Serial	50	993	0.4148	0.98	Yes
Approximate Entropy	50	993	0.5022	1.00	Yes
Linear Complexity	50	993	0.4994	1.00	Yes

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VII. PERFORMANCE COMPARISON

The proposed SDV PUF is compared with several existing designs in Table XII, including Hybrid RO-PUF,
Low-Hardware CRO-PUF, XNOR-XOR RO PUF, MCRO PUF, ISSCC 2020 PUF, XOR RO-PUF, PRO PUF, and Kernel-based CRO. The proposed structure uses static D-Flip-Flop delay counters as the entropy source and is developed in 180 nm CMOS technology aimed at ASIC. It operates consistently between 0 to 125 °C temperature and 1.62 to 1.98 V supply voltage, and it achieves a very low power range of 1.02 nW–49.27 nW and compact area use (376–3943 cells per bit).

TABLE XII
COMPARISON OF PUF DESIGNS ACROSS DIFFERENT CRO IMPLEMENTATIONS

PUF	**Hybrid [11]**	**Low HW [16]**	**XNOR-XOR ROPUF [17]**	**MCRO PUF [19]**	**ISSCC 20 [46]**	**XOR [47]**	**PRO [48]**	**Kernel-based CRO [2]**	**This work (SDV)**
Technology (nm)	28	-	180	-	28	45	40	22	180
Target Hardware	FPGA	FPGA	ASIC	FPGA	ASIC	FPGA	FPGA	ASIC	ASIC
Entropy source	Logic gates	XOR/XNOR	RO frequency	-	NAND	XOR	Switch boxes	Logic gates	Static D-Flip Flop delay counter
Max. No. CRPs	8M	214	-	10K	-	28	-	232	10K(for each

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									configuration)
Challenge size (bit)	18	14	Variable	[n×(k−1)]!	n	8	25	32	Variable
PUF size (configurations)	128 HC-ROs	-	Depends on RO count	16	-	256	128,256,512	-	8~64
Power	119.4mW	125mW	4.447 mW	-	378.9mW	-	-	16.8mW	1.02nW~49.27nW
Area/Bit (cell count)	896	352	-	210Slices	-	2048	-	128	376~3943
Timing	0.184ns	-	100 MHz	-	-	-	0.551ns	-	+0.004ps slack (Post routing)
Temperature (°C)	5~70	-	-	0~80	-40~150	0~60	-40~80	25~70	0~125
Voltage (V)	1.16~1.24	-	-	-	0.8~0.99	0.8~1.2	0.9~1.3	0.72~0.88	1.62~1.98
IntraHD (%)	1.78	1.67	0.14	-	10.5	1.71	1.99	9.95	23.95~42.49
Uniformity (%)	50.36	49.95	67%	-	-	-	49.74	49.42	34.90~55.20
Uniqueness (%)	46.76	50.1	49.62%	48.49	49.78	48.438	48.63	45.5	27.98~55.92
Size of Trained CRPs	280	13K	-	-	-	-	20K	1M	7K
ML prediction Accuracy (%)	90	70	-	-	-	-	72	65.3	69.66~72.93
Unique chip feature	Delay variation in hybrid logic gates	XOR/XNOR Logic gate process variation	-	-	Threshold voltage variation between NAND cells	Delay variation in LUT	Programmable Interconnect Points	Delay variation on interconnect paths	Delay variation in data path on chip layout

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Strong chip differentiation is indicated by an intra-die hamming distance variation of around 42.49% and uniqueness ranging from 27.08%to 54.92%. The proposed layout exhibits strong resistance to modelling attacks, maintaining competitive uniformity (34,9-55.2%) and ML prediction accuracy (~70%) compared to other PUFs. Compared to other delay-based PUFs, the proposed PUF has less power and timing overhead, and the unique chip feature is derived from the delay variation in the data stream on the chip layout.

From the comparison, it is noted that the static D-flip flop-based delay counter has a significant advantage for the proposed PUF over other CRO PUF. Especially, it is suitable for scalable delay variant security applications such as anticounterfeiting and IC authentication. It is

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used to improve digital precise timing. It gives stability against temperature and voltage fluctuation. It targets power constrained environments.

VIII. CONCLUSION WITH FUTURE WORK

The proposed SDV PUF is developed using 180 nm CMOS technology by focusing on different data paths among chip cores, which leads to lower power consumption and improved timing efficiency. The design is a reliable PUF with a higher number of challenge-response pairs and resilient performance since it complies with ISO/IEC 20897-1 and 20897-2 security criteria. With uniformity ranging from 34.9% to 55.2%, uniqueness ranging from 27.98% to 55.92%, and an average ML prediction accuracy of over 50-70%, it demonstrates moderate resistance to modelling attacks. Hence, it is a weak PUF, but provide application for anti-counterfeiting through improving the unique signature on chip layouts. Future studies will incorporate more nonlinear functions, error correction, and aging compensation solutions. It can further increase reliability, uniqueness, and uniformity for long-term secure hardware applications.

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FUNDING

The authors received no funding for this work.